Deploying Web App on ECS - Step 8: Enable Service Auto Scaling (Scheduled Scaling)

To Begin with the Lab

Summary of the Lab

In this step, **Amazon ECS Service Auto Scaling** was configured using **scheduled scaling** so the application could increase and decrease task count at specific times. The lab first set minimum and maximum task limits for the service, then created a scheduled scale-out action that increased the desired count from 2 to 4 tasks. After that, a scheduled scale-in action was created so the service could return from 4 tasks back to 2 tasks later. The running tasks were verified through the service details and the **Application Load Balancer** target group, which automatically registered the new tasks. This showed that scheduled scaling can manage predictable traffic changes without needing manual intervention or load-based alarms.

Understanding Scheduled Scaling in Amazon ECS

Scheduled scaling in **Amazon ECS** lets you increase or decrease the number of running tasks at predefined times. It exists for workloads that follow a predictable pattern, such as business hours, weekends, or nightly low-traffic periods. Instead of watching CPU or memory like target tracking or step scaling, scheduled scaling uses time-based actions to scale out and scale in automatically. For example, you can increase tasks before peak usage starts and reduce them again when traffic becomes low, which keeps the app available while saving cost.

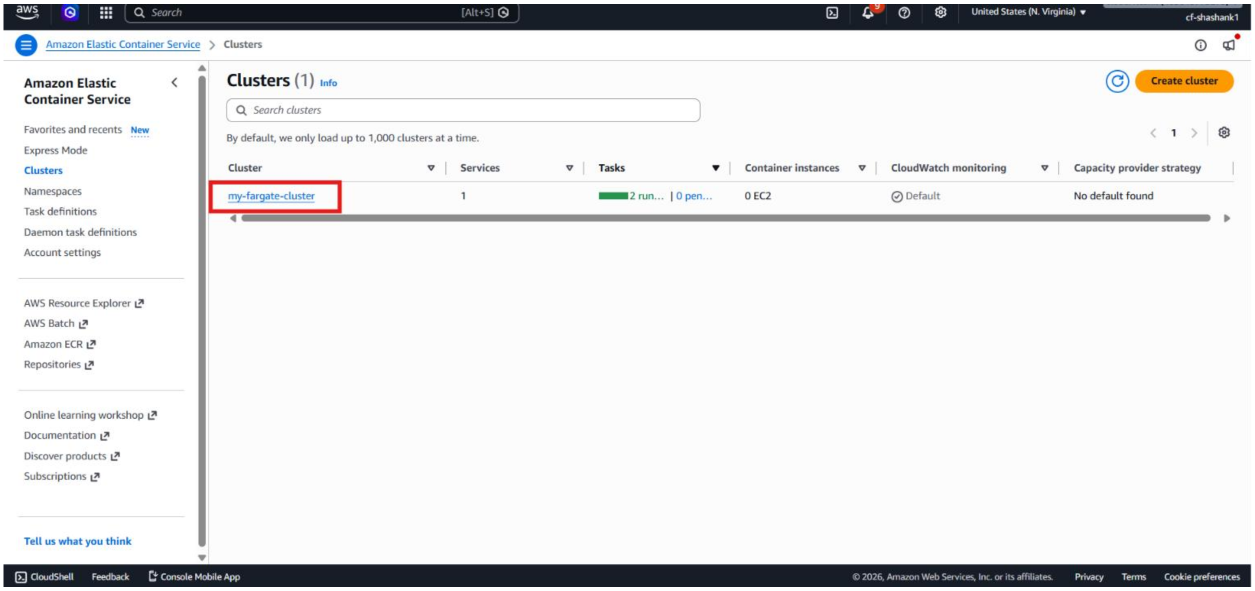
Example:

A PHP web app is running with 2 tasks during normal hours. At 22:55, a scheduled action increases the service to 4 tasks, and later at 23:00 another scheduled action reduces it back to 2 tasks. The **Application Load Balancer** automatically keeps routing traffic to the healthy tasks during both changes.

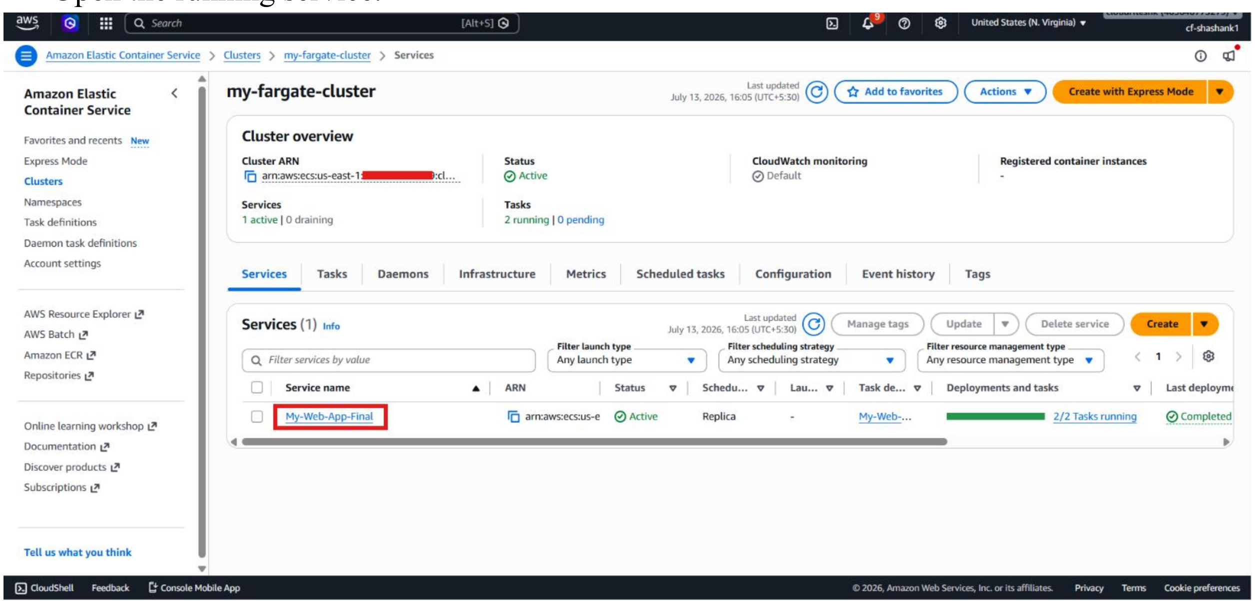
Lab Steps

Step 1 – Open ECS service auto scaling

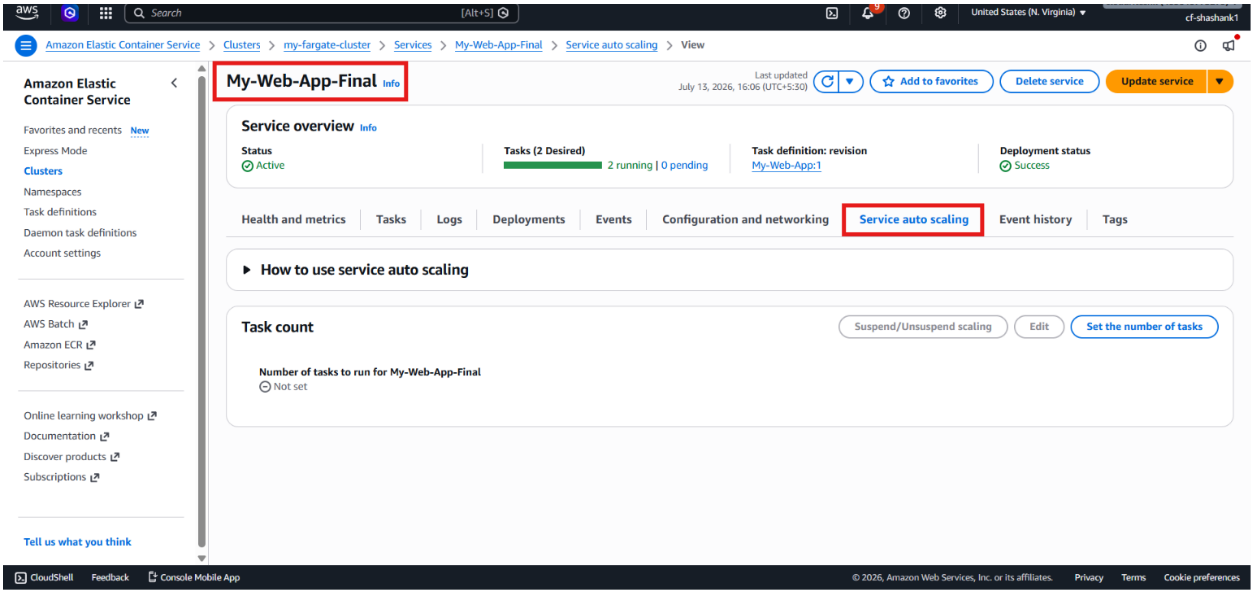
- Sign in to the **AWS Management Console**.
- Open **Amazon ECS**.
- Open the Fargate cluster.



- Open the running service.

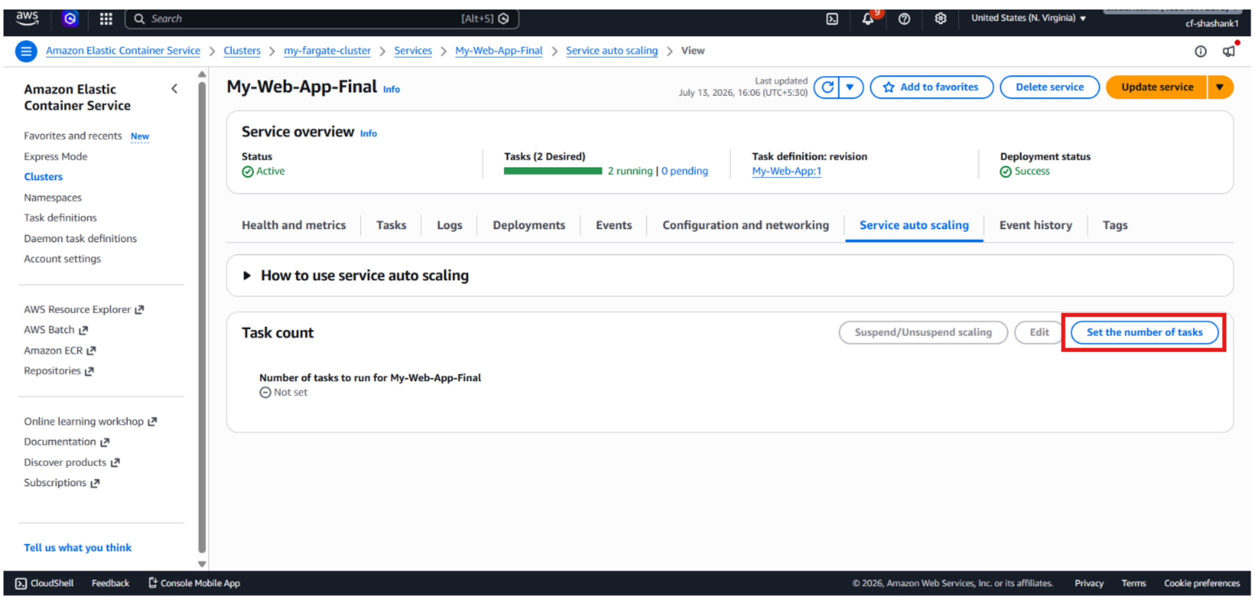


- Go to **Service auto scaling**.

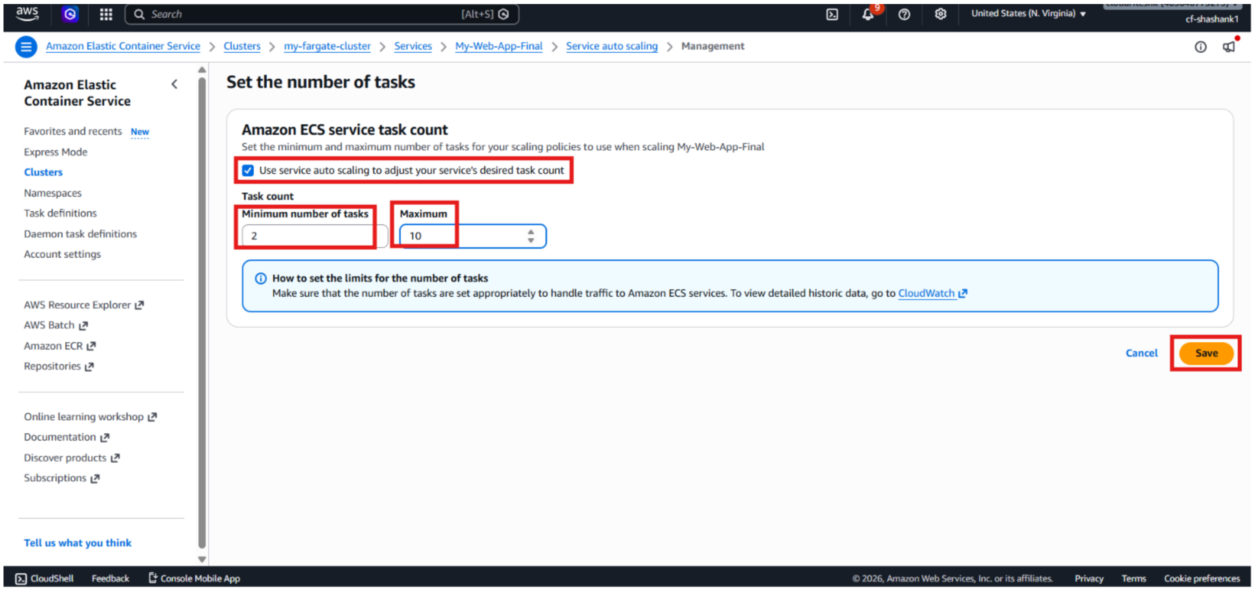


Step 2 – Set the desired task limits

- Click **Set the number of tasks**.

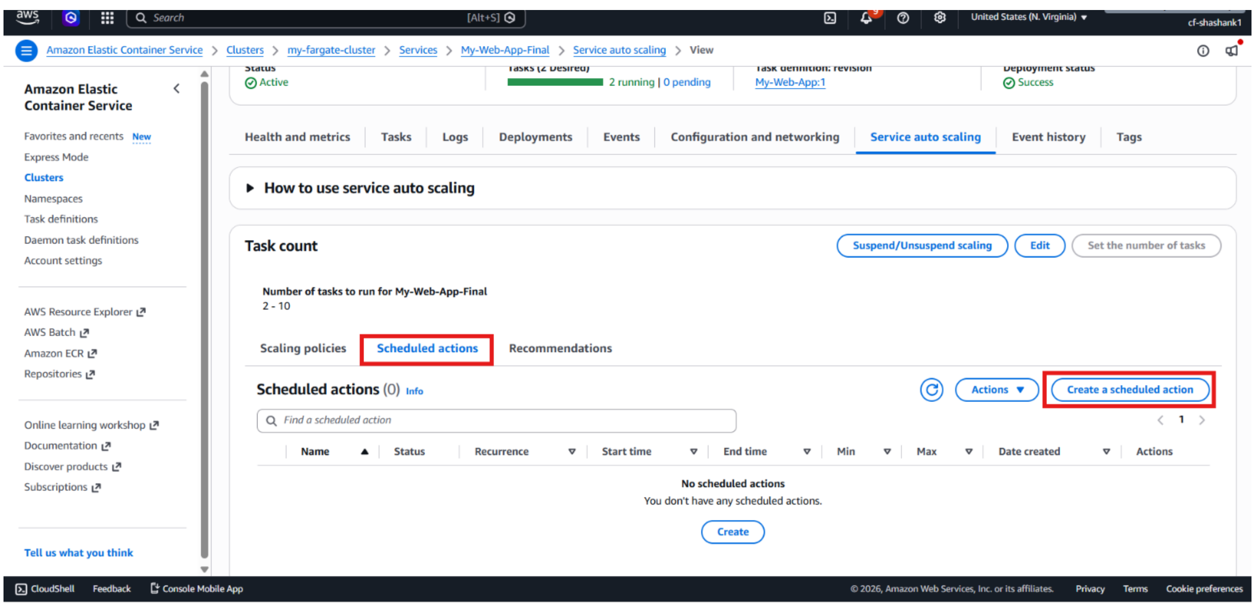


- Set the minimum task count to 2.
- Set the maximum task count to 10.
- Save the task limits.

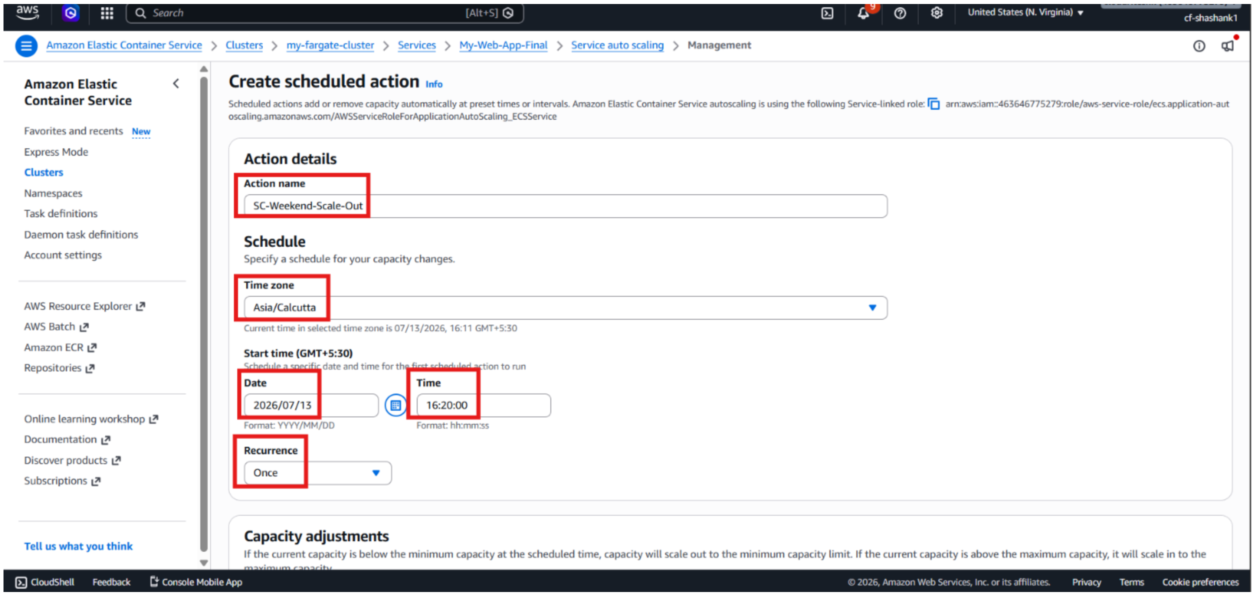


Step 3 – Create the scale-out scheduled action

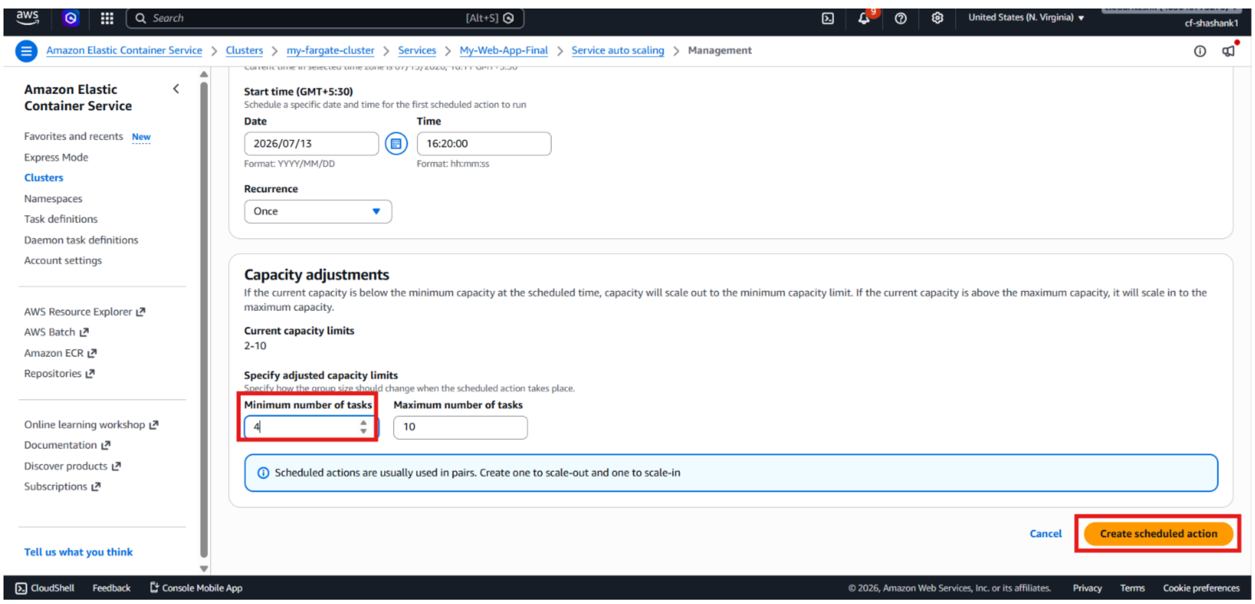
- Click **Create schedule action**.



- Enter the scale-out policy name: SC-Weekend-Scale-Out
- Set the time zone to **Asia/Calcutta**.
- Choose today's date.
- Set the trigger time for the scale-out action.

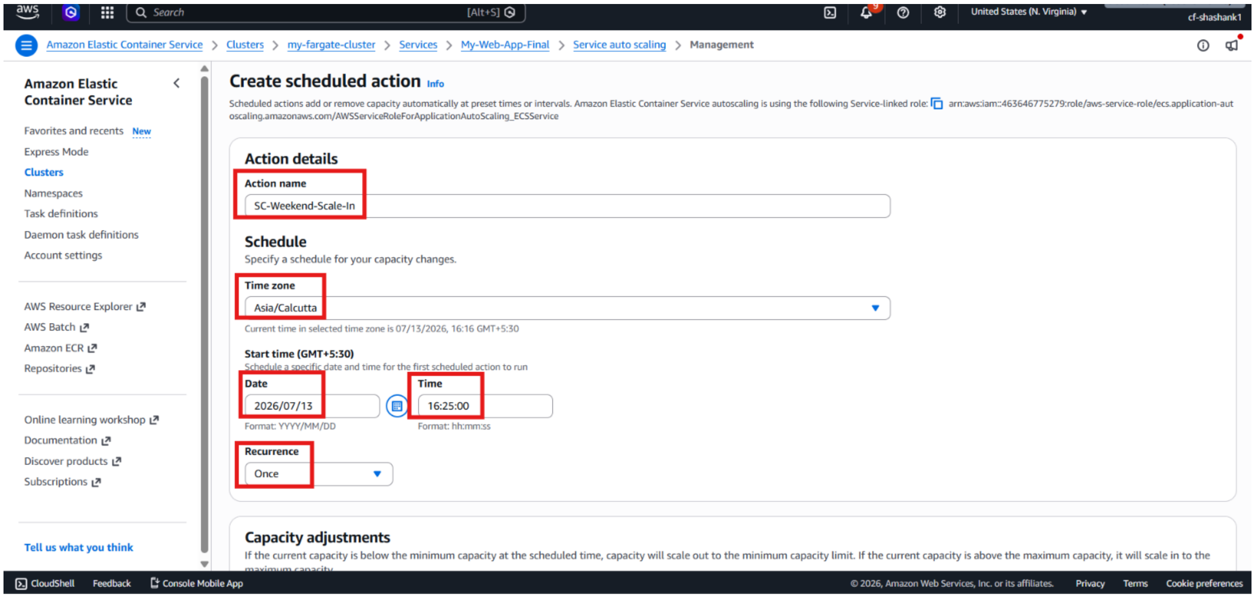


- Set the desired count to increase from 2 to 4.
- Save the scheduled scale-out action.

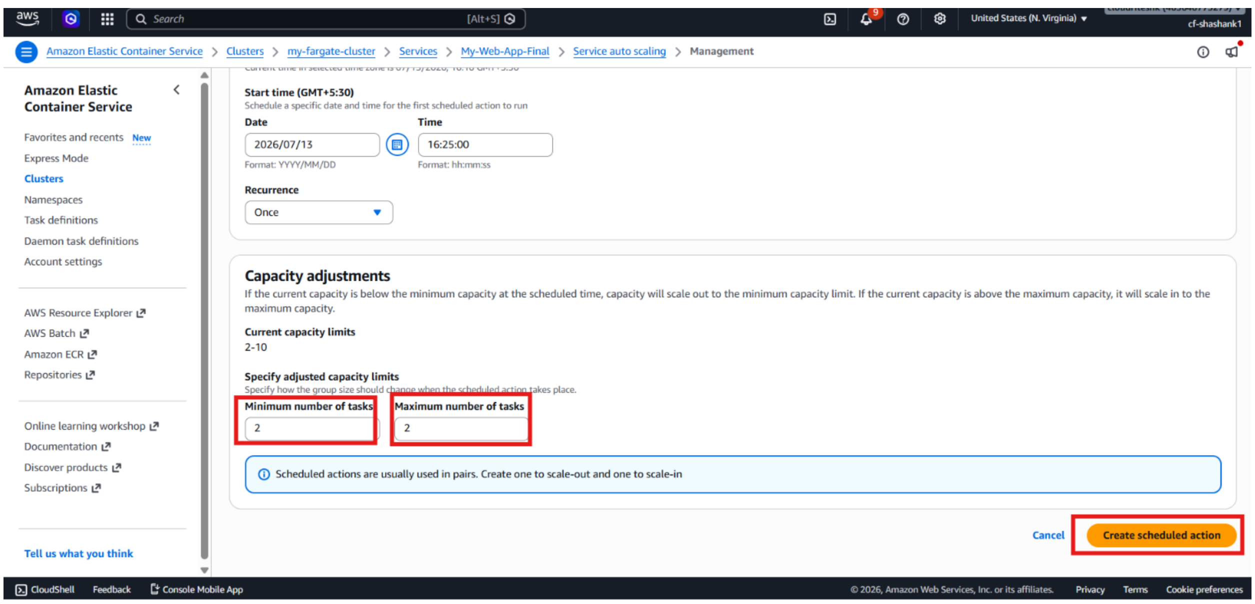


Step 4 – Create the scale-in scheduled action

- Click **Create schedule action** again.
- Enter the scale-in policy name: SC-Weekend-Scale-In
- Set the same time zone and date.
- Choose a later trigger time for scale-in.

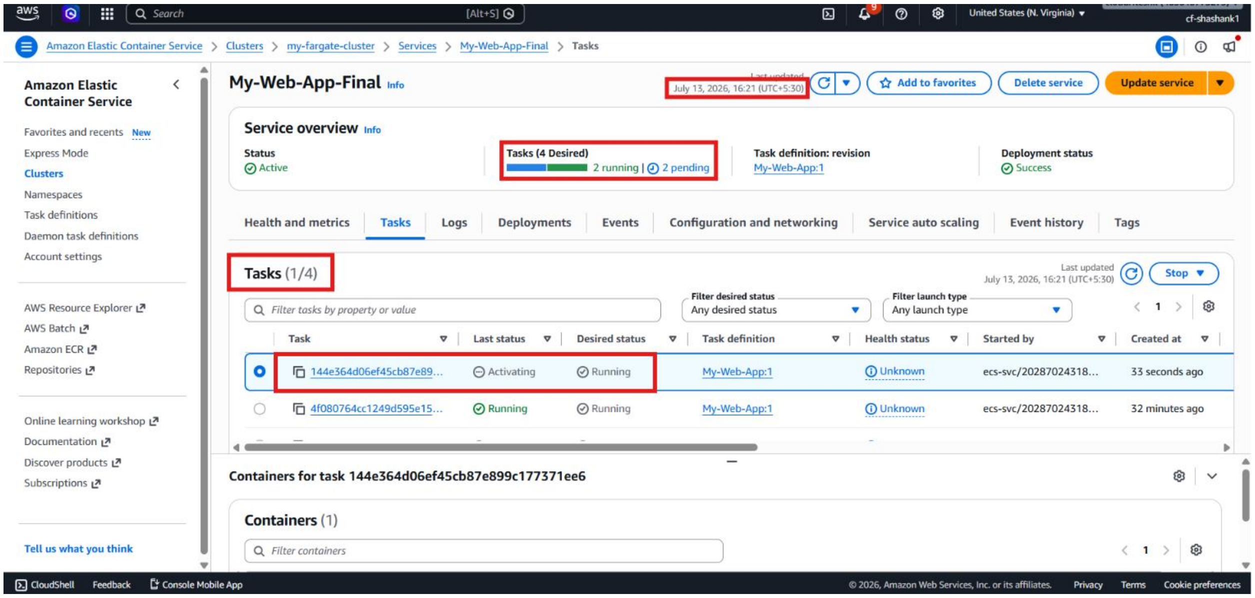


- Set the desired count to return from 4 to 2.
- Set the Max count from 10 to 2 for proper verification.
- Save the scheduled scale-in action.

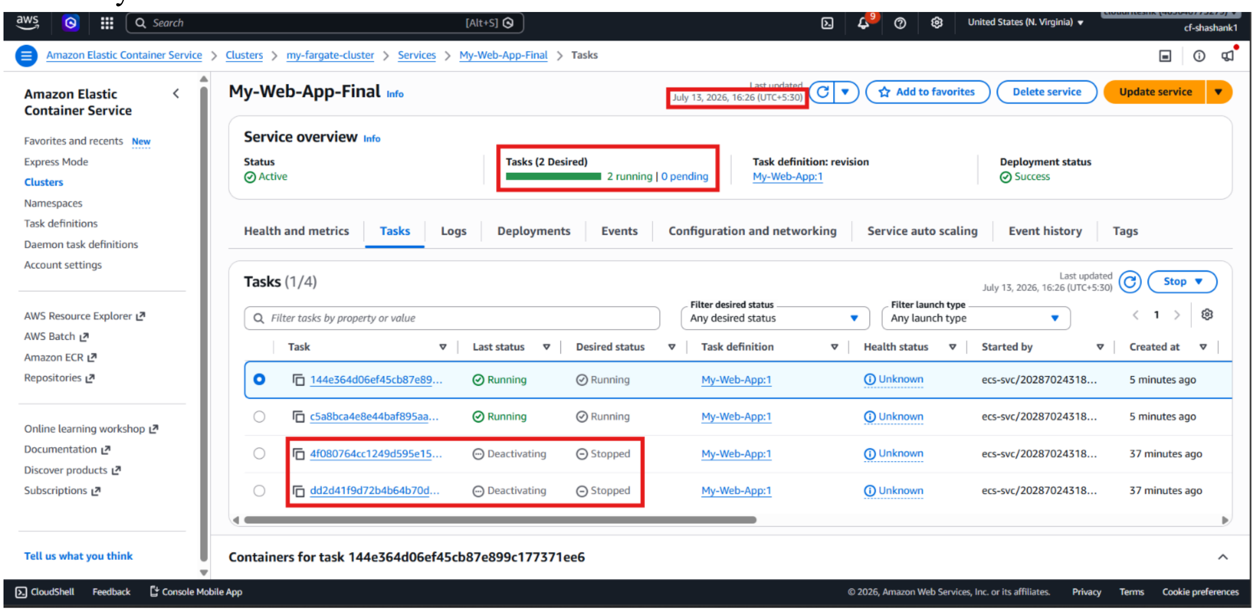


Step 5 – Wait for the scheduled actions to run

- Keep the service running until the scale-out time is reached.
- Verify that the desired task count increases from 2 to 4.

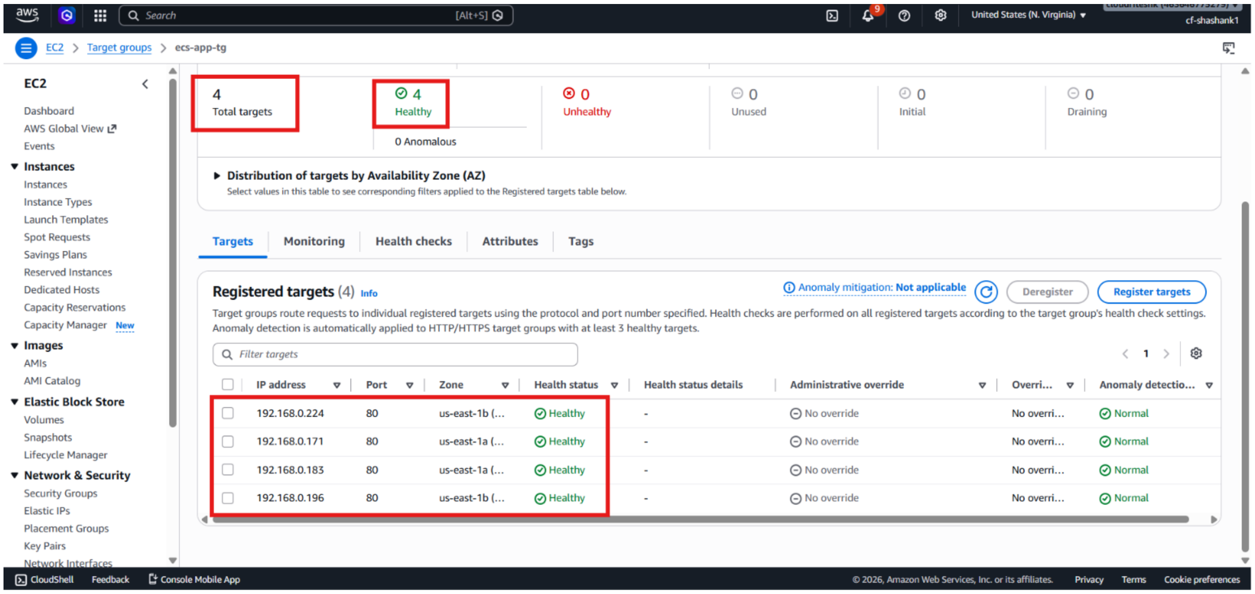


- Wait until the scale-in time is reached.
- Verify that the desired task count decreases back to 2.



Step 6 – Verify load balancer and target registration

- Open the application through the **Application Load Balancer** DNS name.
- Refresh the application and confirm that the container IDs change across tasks.
- Open the **EC2** target group view.
- Confirm that the newly created tasks are registered automatically as healthy targets.



Verification

- Verified that **Amazon ECS Service Auto Scaling** was enabled.
- Verified that the minimum and maximum task limits were set correctly.
- Verified that the scale-out scheduled action increased the service from 2 tasks to 4 tasks.
- Verified that the scale-in scheduled action reduced the service from 4 tasks back to 2 tasks.
- Verified that the **Application Load Balancer** kept routing traffic during scaling.
- Verified that the ECS tasks were automatically registered in the target group.
- Verified that scheduled scaling worked correctly for both scale-out and scale-in actions.